S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

RiskPulse

A real-time AI/NLP risk engine that reads news and social posts, turns each into a structured risk signal, and acts on it: a tactical index rebalancer and a wholesale-bank stress test.



Anish Karthik Ayyagari

IIT Kharagpur

Risk desks see the news; they cannot score all of it

The problem

Thousands of headlines and posts a day, in free text

Analysts triage by hand; stress tests run on fixed schedules

A static shock ignores what an event actually does to rates, spreads and sectors

1

Read

News (GDELT, RSS) and social posts, deduplicated into stories

2

Score

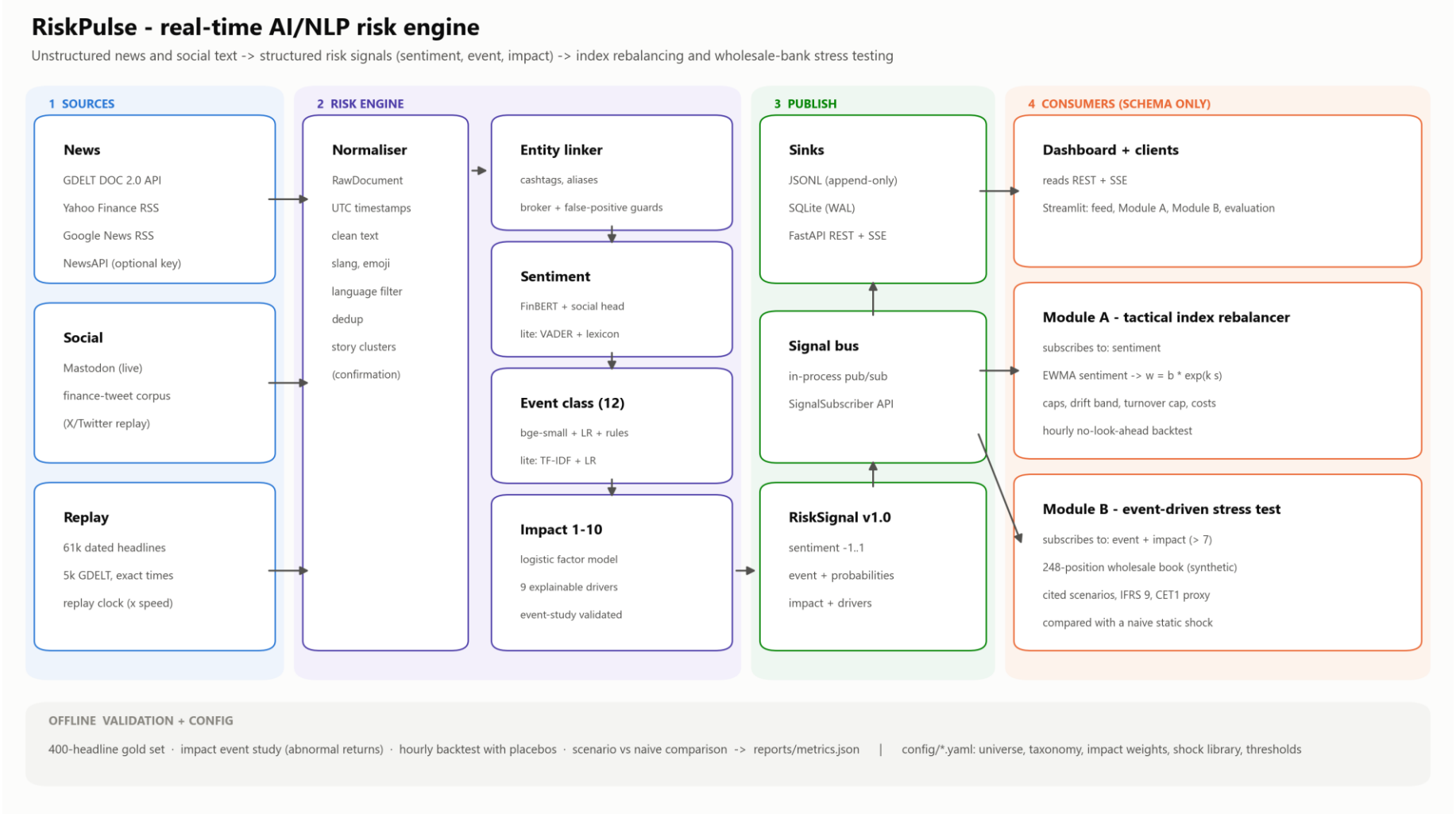
Sentiment −1…1, one of 12 risk events, impact 1–10 with its drivers

3

Act

Module A tilts index weights; Module B stress-tests a $8.5bn book

One engine, decoupled consumers, one versioned signal



Built for a risk desk: explainable, testable, offline-first

Hybrid NLP

FinBERT + a tweet-trained head blended by source; 12-class event model with high-precision rules

Explainable impact

Logistic factor model: 9 drivers stored on every signal, validated by an event study

Decoupled design

Modules read only the RiskSignal schema via in-process, SQLite or SSE subscribers

Module A

Sentiment tilt with caps, drift band, turnover cap and costs; hourly no-look-ahead backtest

Module B

Fires only on reported adverse events; cited scenarios scaled by impact, IFRS 9 staging, Basel CRE20 weights, CET1 proxy

Runs anywhere

Lite tier without torch, offline replay demo, 111 tests, CI on Python 3.11

Measured against naive baselines on held-out data

0.69 vs 0.46

sentiment macro-F1, 200 held-out headlines (naive: VADER)

0.65 vs 0.51

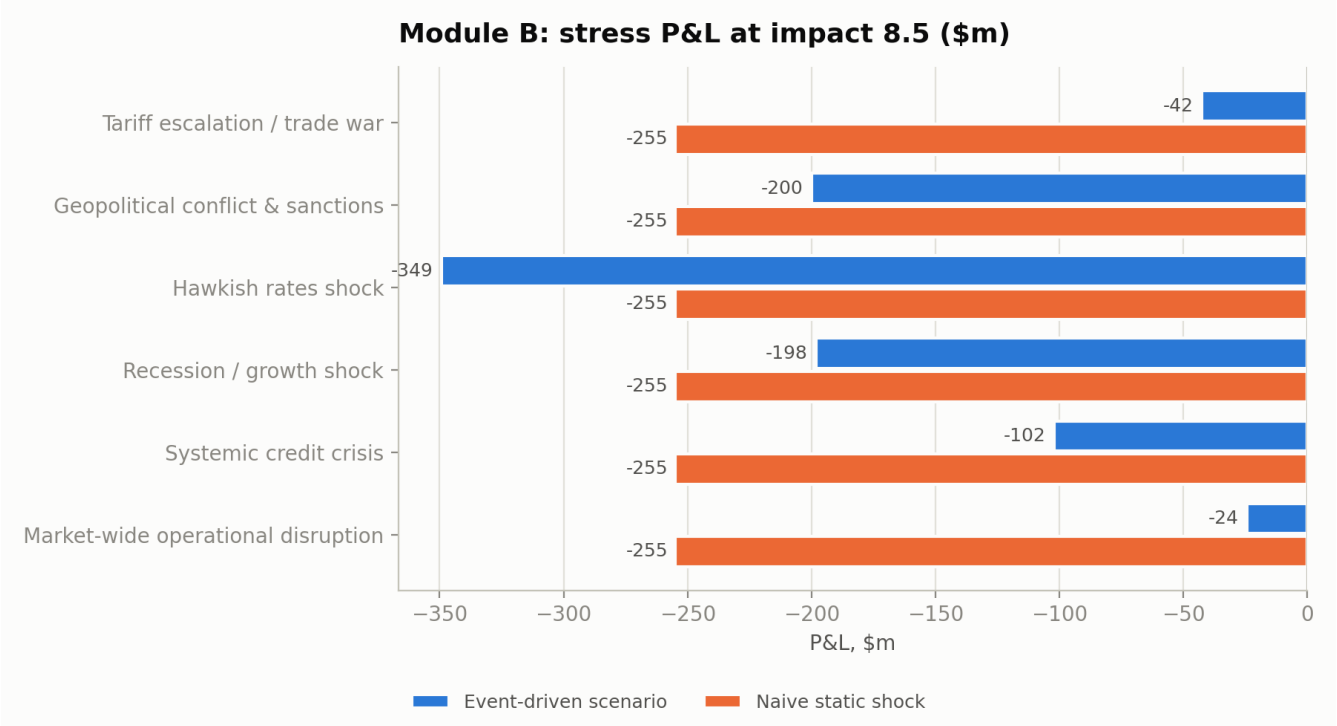
event macro-F1 on the same headlines (naive: keyword rules)

0.09 vs 0.03

impact vs S&P 500 moves, Spearman (naive: |sentiment|)

4 / 6 scenarios

where a naive static stress test gets the rates P&L sign wrong



Headline to stress-test result: 0.2 s on a laptop CPU, vs an overnight batch run (assumed ~24 h).

Honest result: with strict timing, sentiment tilts show no next-day edge (IC -0.004); a first backtest showing IR 3.7 was a look-ahead leak we traced and fixed.

From headline to capital impact in seconds

Risk managers

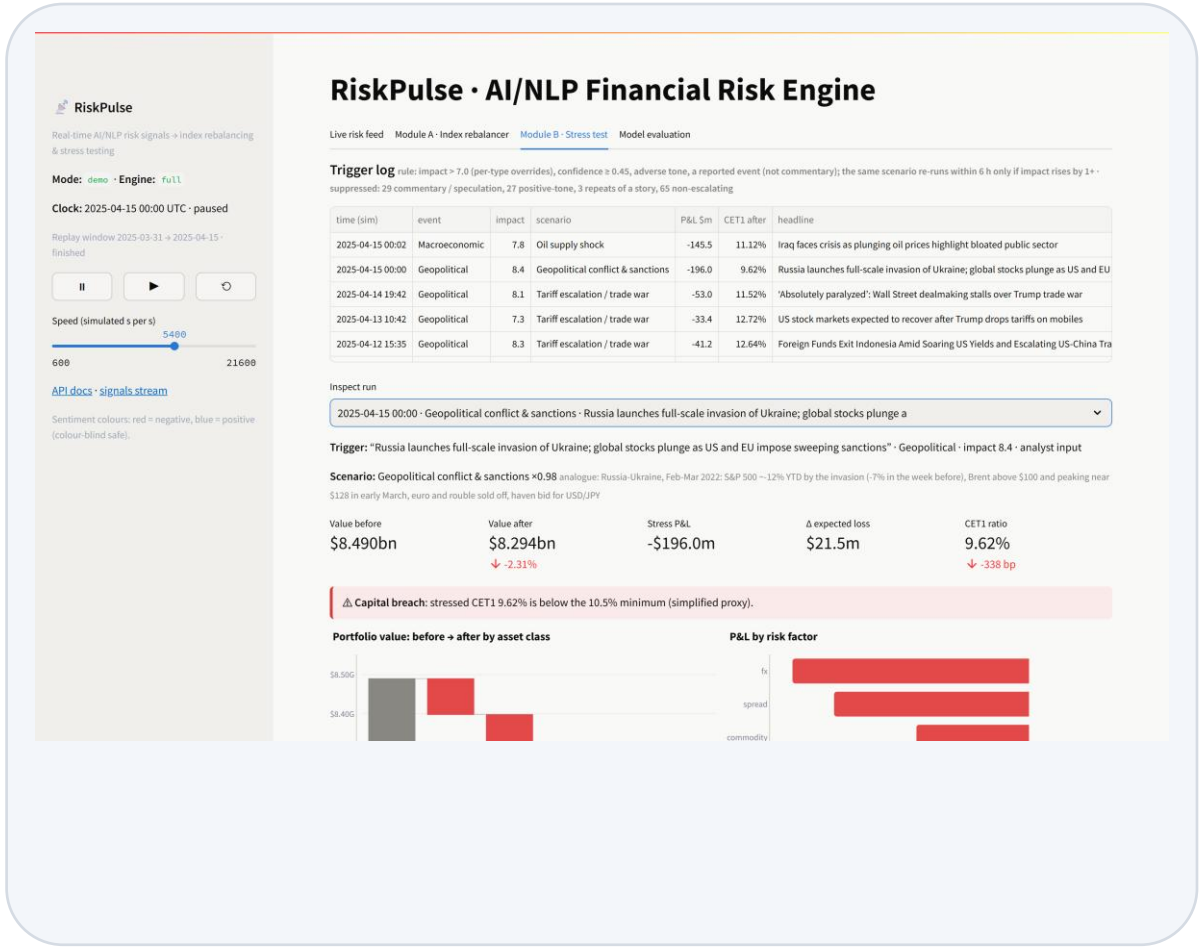
The right stress test runs when an adverse event is reported (not on commentary), with CET1 headroom and the positions that drive the loss

Portfolio managers

Index tilts react to company news within limits, every change linked to its headline

Model risk & audit

Every score carries its drivers, schema and model version; results are reproducible from the repo



What was hard, what is missing, what comes next

Challenges we solved

A backtest information ratio of 3.7 was a look-ahead leak: traced and fixed

97% of 61,399 headlines carry a date, not a time: used only after the day ends

Questions and opinion pieces fired stress tests: now filtered (29 skipped in the replay)

Story clusters swallowed unrelated news: now anchored on the first report

Limitations

Day-level history, so fast intraday reactions are untested

16 mega-cap names and two years of data

Gold labels from one annotator (AI-proposed, author-reviewed)

Sensitivity-based revaluation; simplified CET1 proxy

Next steps

Intraday timestamped feeds to test fast reactions

Multilingual sources and target-aware sentiment

Kafka bus, feature store, model registry

Full revaluation library and regulatory scenarios



Open source (MIT) · public and synthetic data only · github repo and demo video in the README